FastSpeech 2 Text-to-Speech for English and Telugu

Technical Report

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Abstract

We trained two separate FastSpeech 2 acoustic models, one for English (Svarah) and one for Telugu (IndicTTS), each on about 5 hours of speech with 1 hour of non-overlapping validation data. We used the PyTorch implementation of ming024 as the base, the Montreal Forced Aligner for phoneme durations, and the pretrained universal HiFi-GAN vocoder. After 20,000 training steps the Telugu model reaches a validation loss of about 2.23 and the English model about 3.95. The gap is explained mainly by data quality: Svarah is a multi-speaker ASR corpus that we had to treat as a single speaker. All ten test sentences were synthesised intelligibly. The main failure we found is the English front-end, which gives the noun pronunciation of *conduct* in both E2a and E2b. This report describes the implementation, data, training behaviour, per-sentence results, a comparison of the two languages, and how to reproduce everything.

Contents

1 Implementation approach **3**

1.1 Components taken from the reference implementation . . . 3

1.2 Modifications for English and Telugu . . . 3

2 Data, preprocessing and training configuration **4**

2.1 Datasets and subsampling . . . 4

2.2 Alignment and feature extraction . . . 4

2.3 Training configuration . . . 4

3 Training and validation loss **5**

4 Test sentences and speech quality **7**

4.1 English . . . 7

4.2 Telugu . . . 8

5 Variance-adaptor analysis **9**

5.1 Per-sentence observations . . . 9

5.2 Figures . . . 10

6 English versus Telugu **19**

6.1 Limitations and failures seen on the test sentences . . . 19

6.2 Possible improvements . . . 19

7 Reproducibility **21**

7.1 Environment . . . 21

7.2 Commands . . . 21

7.3 Repository layout . . . 21

1 Implementation approach

We started from the reference repository ming024/FastSpeech2 and kept its architecture and loss. Two copies of the code live in `English/` and `Telugu/`, each with its own `config.yaml`, `preprocess.py`, `train.py`, `inference.py` and `checkpoints/`.

1.1 Components taken from the reference implementation

- **Encoder:** 4 feed-forward Transformer (FFT) blocks, hidden size 256, 2 attention heads, convolution filter size 1024.
- **Variance adaptor:** separate duration, pitch and energy predictors (filter size 256). Duration is predicted in log domain and used by a length regulator to expand phoneme states to frame level. Pitch and energy are *phoneme-level* values, quantised into 256 linear bins and added as learned embeddings.
- **Decoder:** 6 FFT blocks, hidden size 256, followed by a mel-spectrogram linear layer and a convolutional post-net.
- **Loss:** sum of mel loss, post-net mel loss, duration loss (MSE in log domain), pitch loss and energy loss.
- **Vocoder:** pretrained universal HiFi-GAN at 22.05 kHz, used unchanged.

1.2 Modifications for English and Telugu

- **English front-end:** CMU/Librispeech lexicon lookup (ARPAbet) with the `g2p_en` model for out-of-vocabulary words. Numbers, times and dates are verbalised before lookup.
- **Telugu front-end:** Unicode NFC normalisation, removal of ZWJ/ZWNJ, and keeping only the Telugu block (U+0C00–U+0C7F). Every Telugu character is one model symbol, added to `text/symbols.py` with an `@` prefix. A Telugu number verbaliser is applied to digits.
- **Speakers:** English is single-speaker (`spk0`). Telugu is multi-speaker (female and male) with a speaker embedding.
- **Compatibility fixes** for current libraries: updated `librosa` STFT arguments, the `tgt` interface for TextGrids (empty intervals mapped to the silence symbol `sp`), `weights_only=False` and `map_location="cpu"` for `torch.load` so that inference also works without a GPU, and a float32 cast of the losses (a Double/Float dtype error appeared in the backward pass).
- **Inference script:** `inference.py` unzips the vocoder on first use, downloads the required NLTK data, runs the five test sentences and saves audio and the three variance plots.

2 Data, preprocessing and training configuration

2.1 Datasets and subsampling

English uses the Svarah dataset (accented Indian English). The Hugging Face copy has no speaker column, so all utterances were treated as one speaker. **Telugu** uses the IndicTTS Telugu corpus with its two speakers (female and male).

For each language we shuffled with seed 42, kept clips between 1 and 15 s, resampled to 22,050 Hz mono, filled the training set up to 5.0 hours and then the validation set up to 1.0 hour. The two sets never share an utterance. Table 1 summarises the result.

Table 1: Data used for each language.

	English	Telugu
Corpus	Svarah	IndicTTS Telugu
Speakers	1 (assumed)	2 (female, male)
Train / validation target	5.0 h / 1.0 h	5.0 h / 1.0 h
Train / validation utterances	3,185 / 637	2,261 / 452
Aligned TextGrids	3,889 of 3,924	2,713 of 2,718
Mean F0 (std)	176.5 Hz (54.6)	192.8 Hz (43.2)

2.2 Alignment and feature extraction

Phoneme durations come from the Montreal Forced Aligner (MFA 3.4.3).

- **English:** pretrained `english_us_arpa` acoustic model and dictionary.
- **Telugu:** an aligner trained from scratch with a grapheme dictionary (one symbol per Unicode character). To stay inside the 12-hour Kaggle session limit we used 10 monophone and 10 triphone iterations instead of the default schedule.

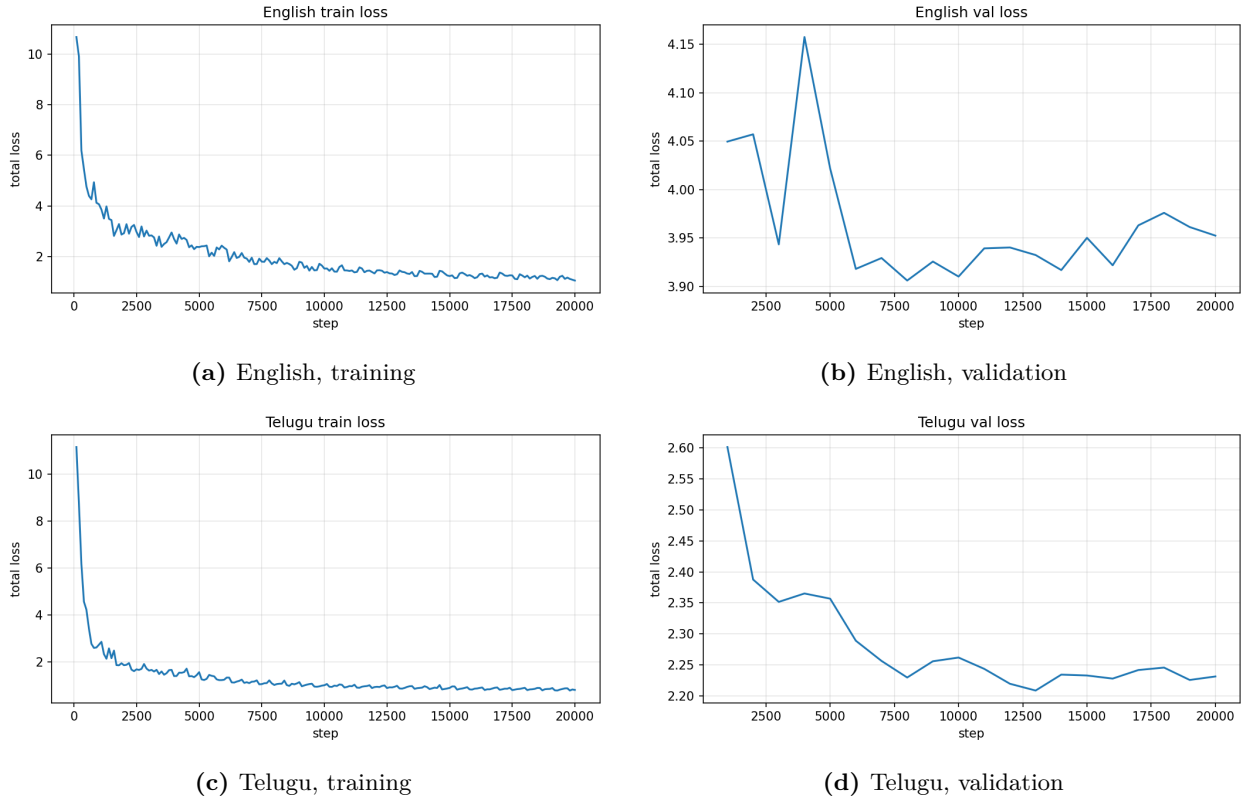
Utterances that could not be aligned (35 English, 5 Telugu) were dropped. Mel-spectrograms, phoneme-level pitch (pyworld) and phoneme-level energy were then computed and normalised with dataset statistics stored in `stats.json`.

2.3 Training configuration

Table 2: Training configuration (identical for both languages).

Total steps	20,000
Batch size	16
Optimiser	Adam, $\beta = (0.9, 0.98)$, $\epsilon = 10^{-9}$
Learning-rate schedule	Transformer warm-up, 4,000 steps
Gradient clipping	1.0
Logging / validation / sample synthesis	every 100 / 1,000 / 1,000 steps
Hardware	one NVIDIA T4 GPU per language
Saved checkpoints	fp16 weights, 68 MB each

3 Training and validation loss



(a) English, training

(b) English, validation

(c) Telugu, training

(d) Telugu, validation

Figure 1: Total loss (sum of mel, post-net, duration, pitch and energy losses). Training loss is logged every 100 steps and validation loss every 1,000 steps.

Table 3: Approximate values read from the loss curves in Figure 1.

	Train (step 100)	Train (step 20k)	Val. minimum	Val. (step 20k)
English	10.7	1.05	3.91 (step 8k)	3.95
Telugu	11.2	0.80	2.21 (step 13k)	2.23

English. The training loss falls quickly from about 10.7 to about 3 in the first 2,000 steps and then declines slowly to about 1.05. It is noisier than the Telugu curve between steps 1,000 and 6,000, with spikes near step 800, 1,300 and 4,000. The validation loss hardly improves: it varies only between 3.9 and 4.16, with one spike at step 4,000, a minimum near step 8,000 and a slight upward drift to 3.95 at the end. A gap of about 3 loss units between training and validation, together with a validation curve that is flat or rising while training loss keeps falling, indicates limited generalisation and mild overfitting to the 5-hour training set. We think the main reason is data quality rather than model size. Svarah is an ASR benchmark with many speakers, microphones and accents that we had to merge into one speaker, so the pitch, energy and duration targets are noisy and cannot be predicted well from text alone.

Telugu. The training loss drops from about 11 to about 0.8, and the validation loss decreases steadily from 2.60 to about 2.21 by step 13,000, then stays within about ± 0.03 of 2.23. The curves are smoother and the train–validation gap (about 1.4) is much smaller, which fits clean studio recordings from two speakers. Validation changes very little after step 8,000, so more training would bring little gain.

Checkpoint choice. In both runs the validation minimum occurs before step 20,000, so early

stopping near step 8,000 (English) and step 13,000 (Telugu) would have been marginally better. We kept the step-20,000 checkpoint for both models because the difference is small (about 0.04 for English and 0.02 for Telugu) and we did not save intermediate checkpoints.

4 Test sentences and speech quality

Table 4 lists the test sentences. Telugu sentences are given in Latin transliteration here; the original script is used in the synthesis (see `Results/Telugu/frontend_log.json`). The English text front-end converted the numbers in E4 into words ("ten forty five A M", "twelve November twenty twenty six") and the Telugu front-end converted 2026, 12 and 10:45 into Telugu number words. In both E4 and T4 a silence symbol (`sp`) was inserted at the comma.

Table 4: Test sentences and what each one tests. Sequence length is the number of model symbols.

ID	Sentence	Tests	Symbols
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	very long rare word	85
E2a	The orchestra's conduct during the ceremony was praised by the audience.	homograph: *conduct* (noun)	50
E2b	The researchers will conduct an experiment to evaluate the proposed method.	homograph: *conduct* (verb)	56
E3	The rhythm of the orchestra changed unexpectedly.	irregular spelling	37
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	numbers, date, out-of-vocabulary word, comma pause	79
T1	*Neuropharmacological parishodhanalo nootana aushadhala prabhaavaanni vishleshinchaaru.*	long technical word	62
T2a	*Taadupai atani pattu chaala balangaa undi.* (grip)	homograph *pattu*	28
T2b	*Aame pattucheeranu dharinchi vedukaku vachchindi.* (silk)	homograph *pattu*	34
T3	*Puli adavilo nishabdangaa nadichindi.* (The tiger walked silently in the forest.)	short, simple sentence	29
T4	*Ee samaavesham 2026 navambar 12va teedi udayam 10:45 gantalaku praarambhamavutundi, samayaniki siddhangaa unchandi.*	numbers, date, comma pause	99

4.1 English

All five English sentences were judged clear and intelligible when listening to the generated files in `Results/English/test_generated_audios`, with no audible breaks or missing words.

- **E1:** the very long word is produced through the lexicon/g2p path (85 phonemes) and was intelligible.
- **E2a/E2b (homograph):** the front-end output for *conduct* is the *same* phoneme sequence in both sentences (`K AA1 N D AH0 K T`, stress on the first syllable, the noun pronunciation). The verb in E2b should have stress on the second syllable (`K AH0 N D AH1 K T`). This is a failure of the front-end, not of the acoustic model: it looks up the word in a lexicon and takes the first entry without using the part of speech. The two sentences therefore sound alike on this word, and only the surrounding context differs.
- **E3:** *rhythm* was converted to `R AY1 DH EH1 M` and pronounced correctly.
- **E4:** the time and date were read correctly as words, the comma gave a clear pause (about 420 ms), and the made-up word *andaroo* was spelled out by g2p as `AE0 N D AA1 R OW0`.

4.2 Telugu

All five Telugu sentences were clear and intelligible, with stable voice quality. The long technical word in T1, the numbers and date in T4 and the short sentence T3 were all pronounced correctly. For T2a and T2b the word *pattu* has the same spelling and therefore the same symbols in both sentences, so its pronunciation is identical in the two files, which is the correct behaviour for a grapheme-based model since the two meanings are pronounced the same in Telugu. The listening notes come from one listener (the author).

5 Variance-adaptor analysis

For every test sentence we plotted the predicted phoneme duration (ms), phoneme-level pitch F0 (Hz) and phoneme-level energy, using the same symbols as the model input. Telugu symbol labels on the x-axis are not readable in the figures because the default plotting font has no Telugu glyphs, but the order and the number of symbols are correct.

General patterns in English. Durations are mostly between 30 and 130 ms; unstressed vowels (`AH0`, `IH0`) and stops are short, and stressed vowels (`AA1`, `EY1`, `AY1`) and word-final phonemes are longer. Pitch follows a declination trend, starting high near the first content word and ending low (around 120 Hz), with local rises on stressed syllables. Energy is strongly phoneme-dependent: stressed vowels have high values and stops and fricatives such as `T`, `K`, `P`, `S` have values close to zero.

General patterns in Telugu. Durations are between about 10 and 220 ms, with many short phonemes (10–25 ms, such as virama-like symbols and short vowel signs) next to long vowels. Pitch starts high (up to about 300 Hz in T1 and T3) and drifts down to about 150–160 Hz at the end of each sentence, which is the typical declination of Telugu declarative sentences. Energy is highest at the start of the utterance and decreases.

5.1 Per-sentence observations

E1 Long utterance without a comma, so there is no pause in the duration plot (longest bars about 128 ms). Pitch ranges from 100 to 210 Hz with rises at the stressed syllables of the long word (for example `AY1` at about 188 Hz and `OW0 M` at 210 Hz), and falls to about 120 Hz at the end of the sentence. The energy plot has a peak near 97 on the stressed vowel in *establish*.

E2a Pitch is the widest of the English sentences with about 118–247 Hz: a first peak of 247 Hz in *orchestra's*, a gradual decline, a local peak on *praised*, the lowest values (about 120 Hz) on *audience* and a small rise on the final `S`. The longest durations (about 174 ms) are on `AA1` in *was* and on the final `S`, and the largest energy peak (about 107) is also on that `AA1`.

E2b Pitch ranges from 123 to 226 Hz, with peaks on *researchers* (`ER0`), *evaluate* (`EY2`, 226 Hz) and *proposed* (`OW1`, 221 Hz), and a final fall to 123 Hz. Energy peaks on `UW0` in *evaluate* (about 72). Compared with E2a, the symbols of *conduct* are identical, but their durations, pitch and energy differ only because of context, which confirms that the variance predictors do not see a verb/noun distinction.

E3 Shortest English sentence. The long durations are on `AY1` of *rhythm* (about 150 ms), on `EY1` of *changed* (162 ms) and on the final `IY0` (150 ms), which shows phrase-final lengthening. Pitch has a high plateau around 230 Hz on *orchestra* and falls to about 153 Hz at the end. Energy peaks at `AY1` (about 77).

E4 The duration plot has a very long `sp` bar (about 418 ms) at the comma, followed by a long `S` (about 278 ms); a long `T` (about 230 ms) occurs in the date, and the last phoneme is lengthened again. Pitch starts high (up to 267 Hz on *meeting will*), stays near 175–210 Hz through the time and date, drops at the pause, resets to about 237 Hz after the comma and then declines to 118 Hz before a small rise at the end. Energy is almost zero at the pause (about 1.5) and highest on `EH1` in *twelve* (about 41). The model therefore learned phrase boundary, pitch reset and pause behaviour.

T1 A long technical word at the start. Pitch begins very high (up to about 302 Hz in the first symbols), then falls to about 200 Hz and declines slowly to about 158 Hz with several local peaks of 240–260 Hz. The first symbols also have the highest energy (up to 105), which then decreases. The longest duration is about 220 ms.

T2a Pitch ranges from 171 to 284 Hz with a high start and an overall fall; the final symbols are the lowest (about 171–186 Hz). One symbol near the start is very long (about 174 ms), and energy peaks early (about 84).

T2b Pitch is 142–267 Hz and clearly falls near the end of the sentence, from about 230 Hz to about 145 Hz. Energy is highest on the first symbols (about 82) and low in the last third. Compared with T2a, the two files have a similar high-start, low-end shape, even though the sentences differ, as expected for statements with the same intonation.

T3 Shortest Telugu sentence. Pitch has the largest start-up peak (about 335 Hz on the fourth symbol), then falls to about 200 Hz and drifts down to about 150 Hz. Durations are fairly even (the longest is about 139 ms at the end) and energy decreases over the sentence.

T4 The longest sentence (99 symbols) has an `sp` pause of about 463 ms at the comma and nearly zero energy at that point. Pitch ranges from 162 to 265 Hz with many local peaks, resets to about 255 Hz after the pause and then falls again to about 162 Hz. The two long clauses therefore have separate declination contours.

A startup pitch above 250 Hz in several Telugu sentences, compared with the Telugu mean of about 193 Hz, suggests that the model uses a higher-pitched (female) speaker voice for these samples or overshoots at the beginning of the utterance; we did not check this separately.

5.2 Figures

Figures 2–11 show the predicted duration, pitch and energy for each test sentence.

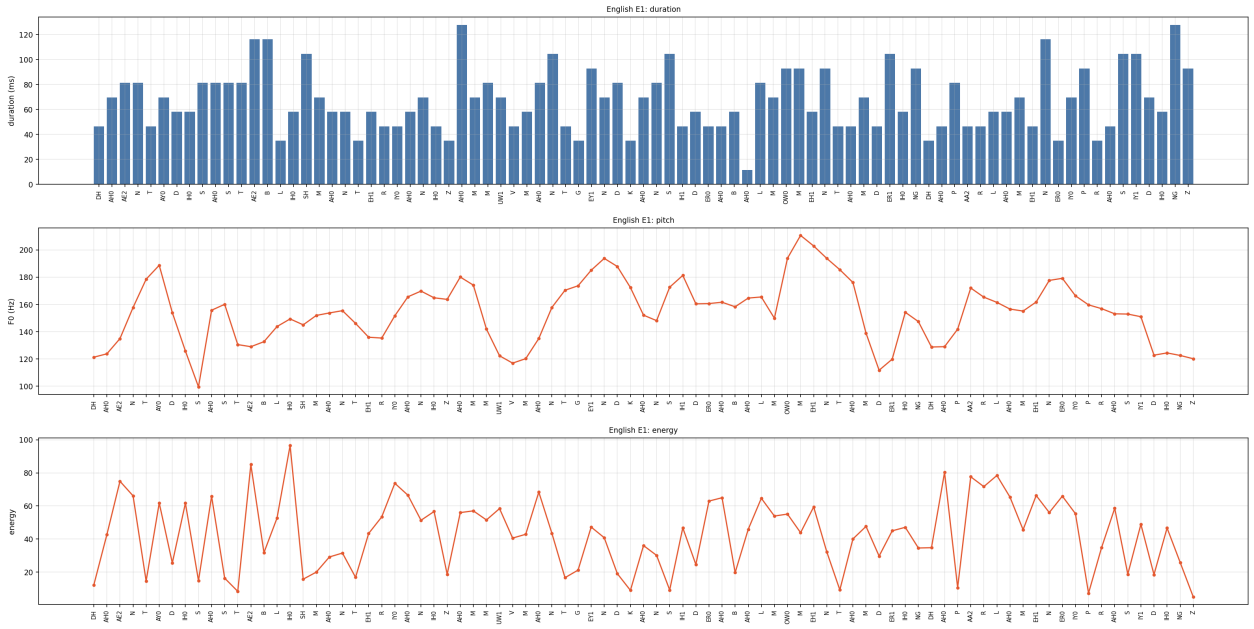


Figure 2: Predicted duration, pitch and energy for E1.

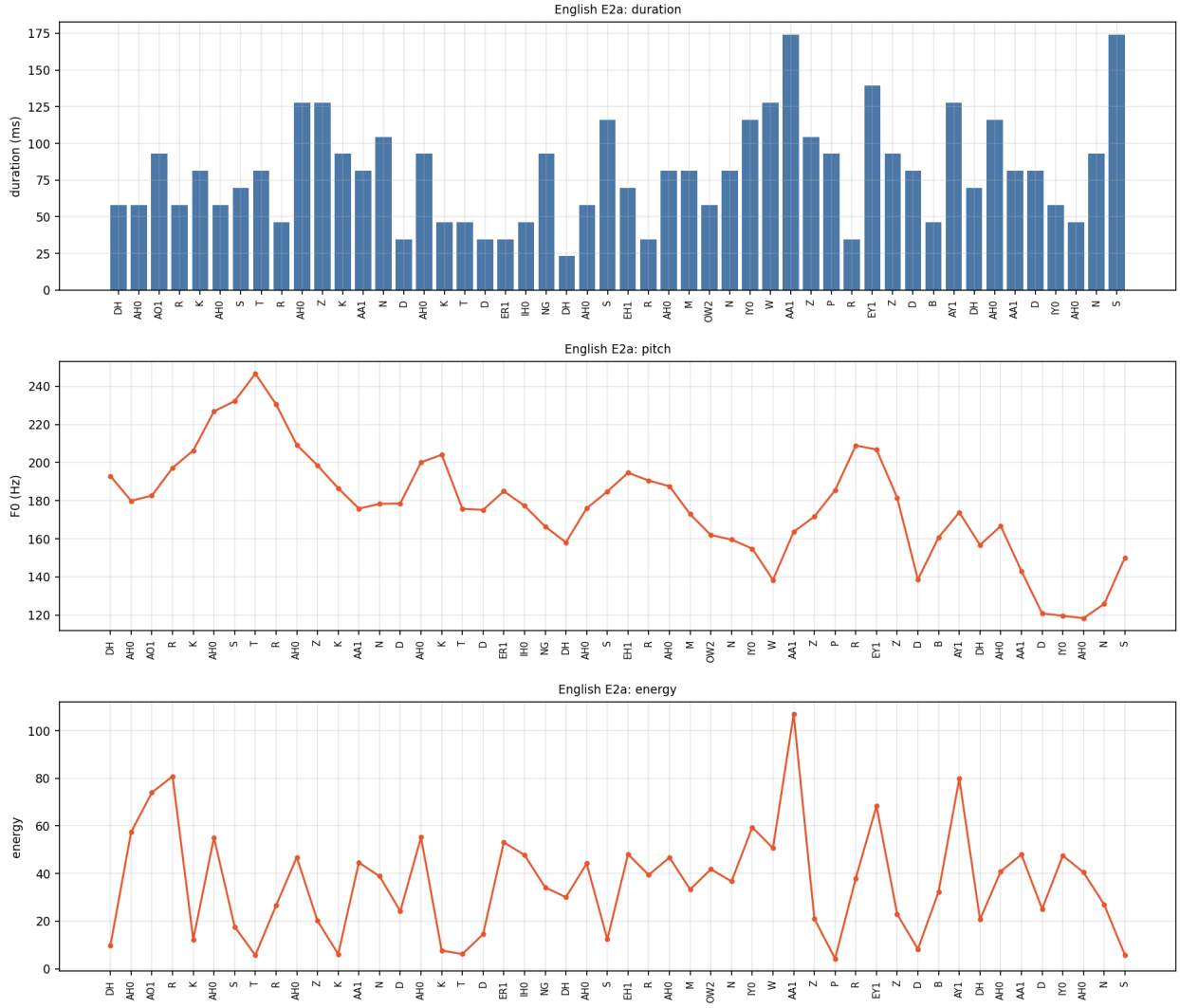


Figure 3: Predicted duration, pitch and energy for E2a.

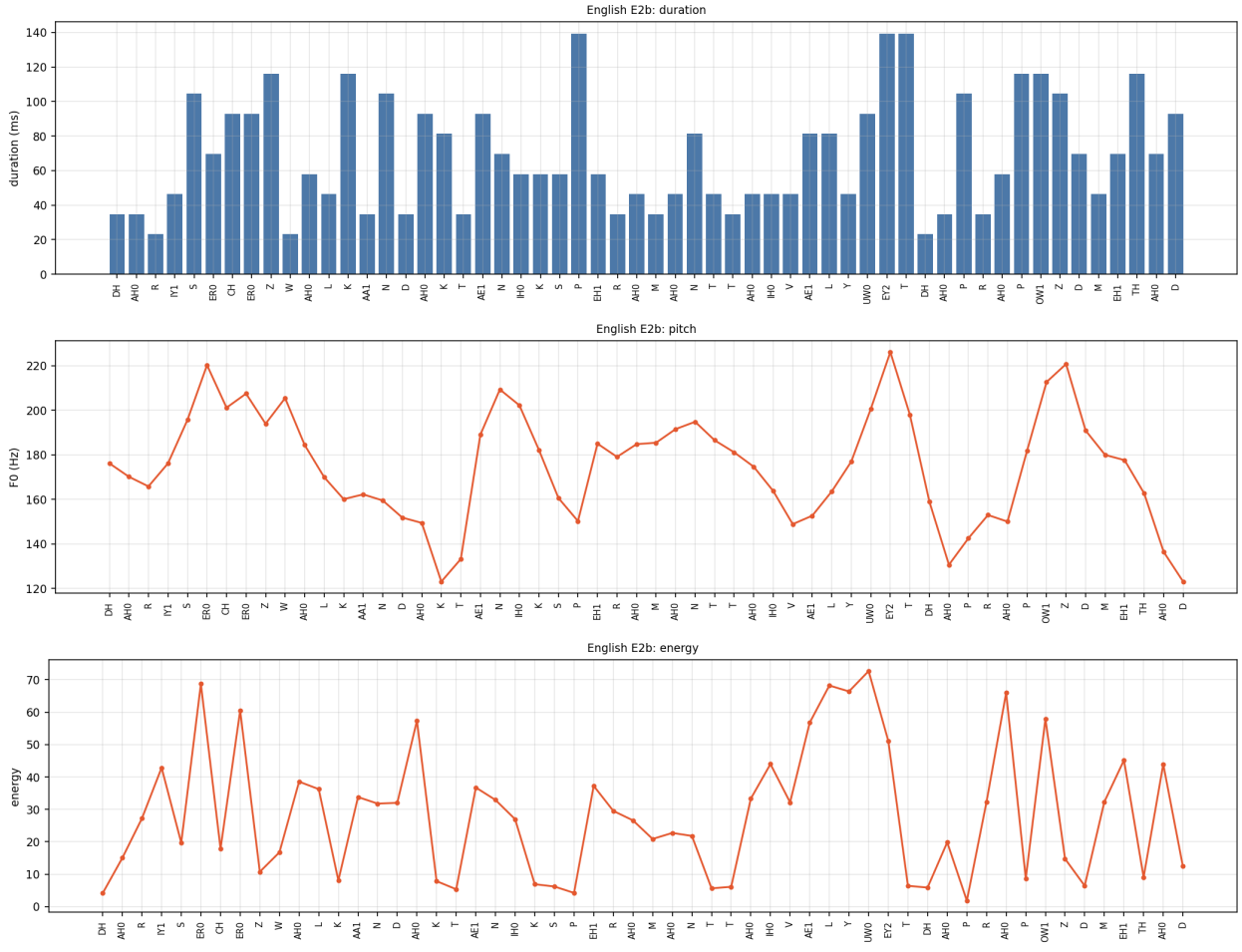


Figure 4: Predicted duration, pitch and energy for E2b.

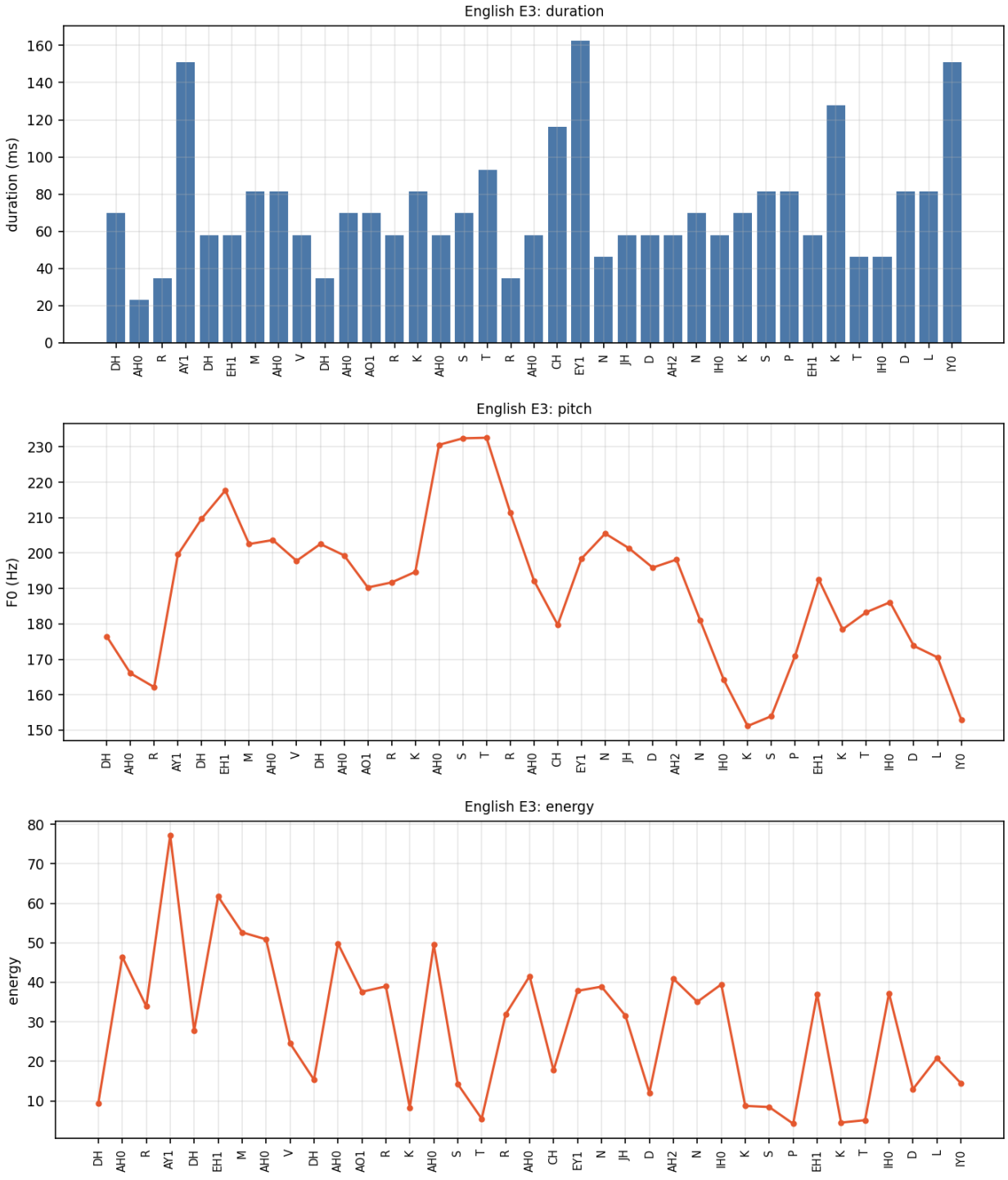


Figure 5: Predicted duration, pitch and energy for E3.

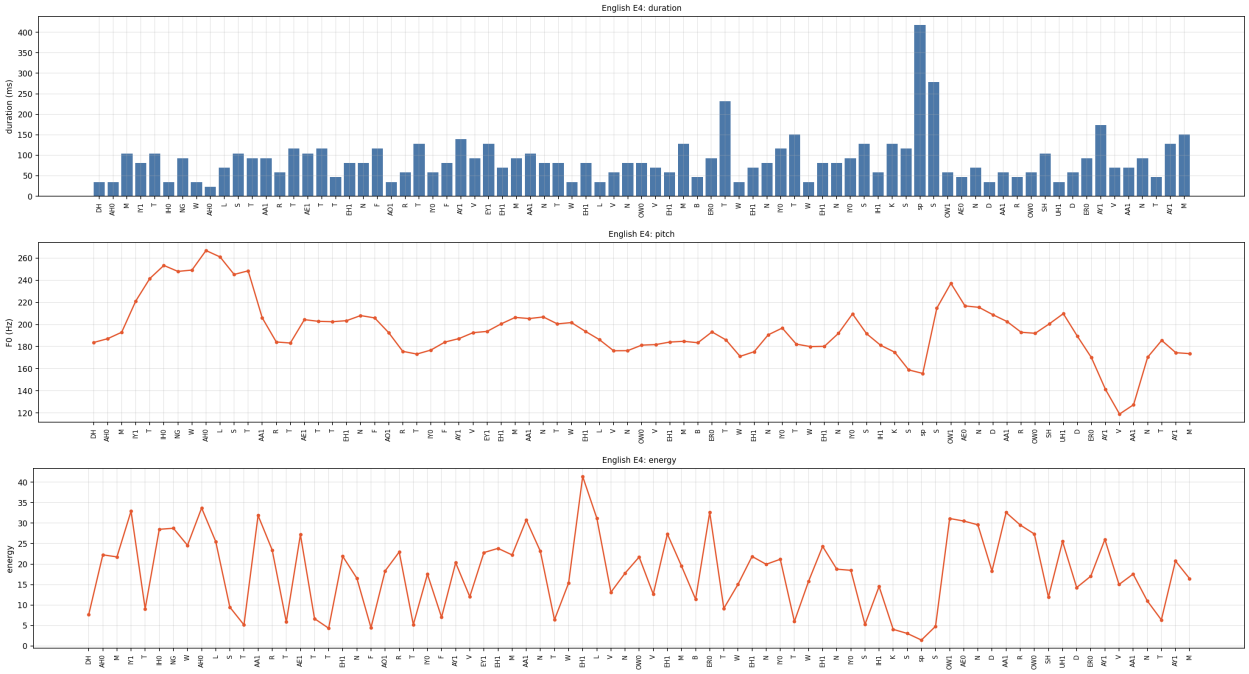


Figure 6: Predicted duration, pitch and energy for E4.

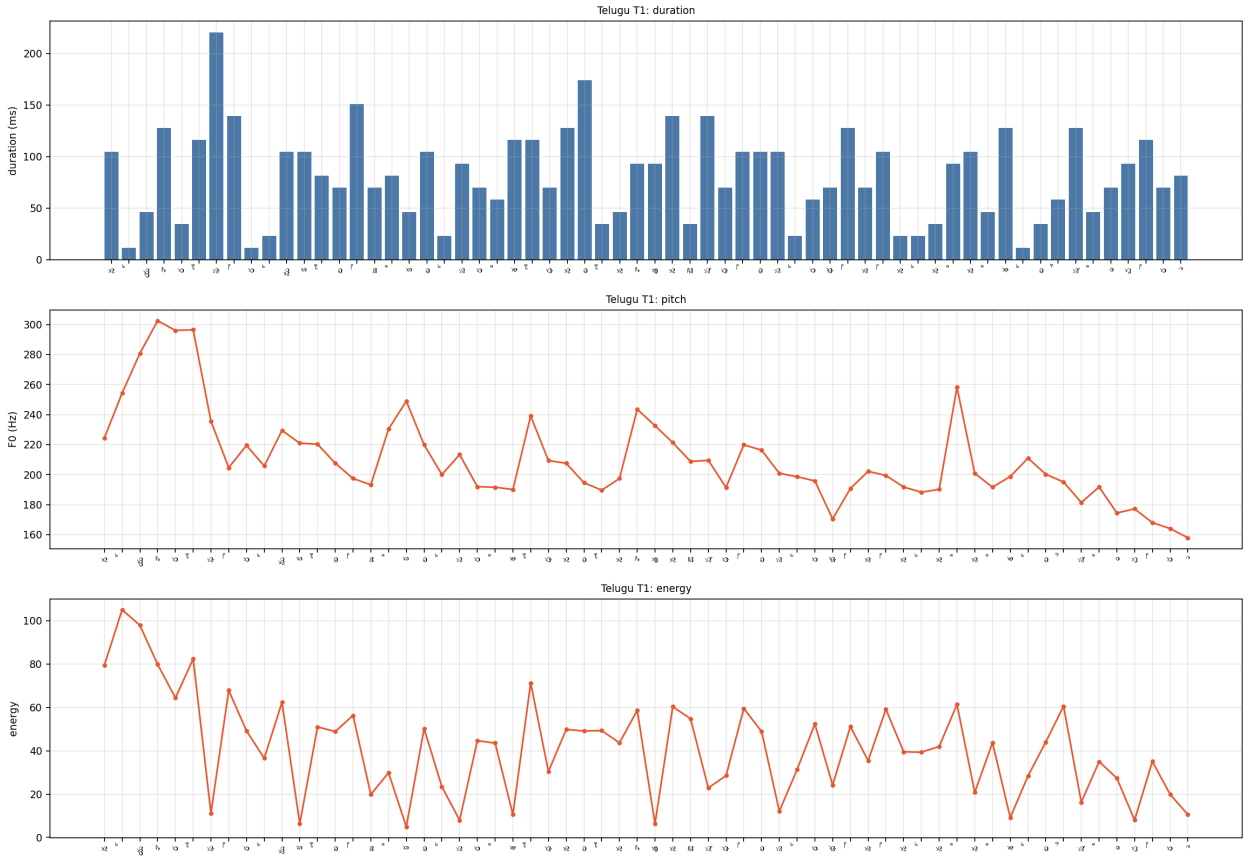


Figure 7: Predicted duration, pitch and energy for T1.

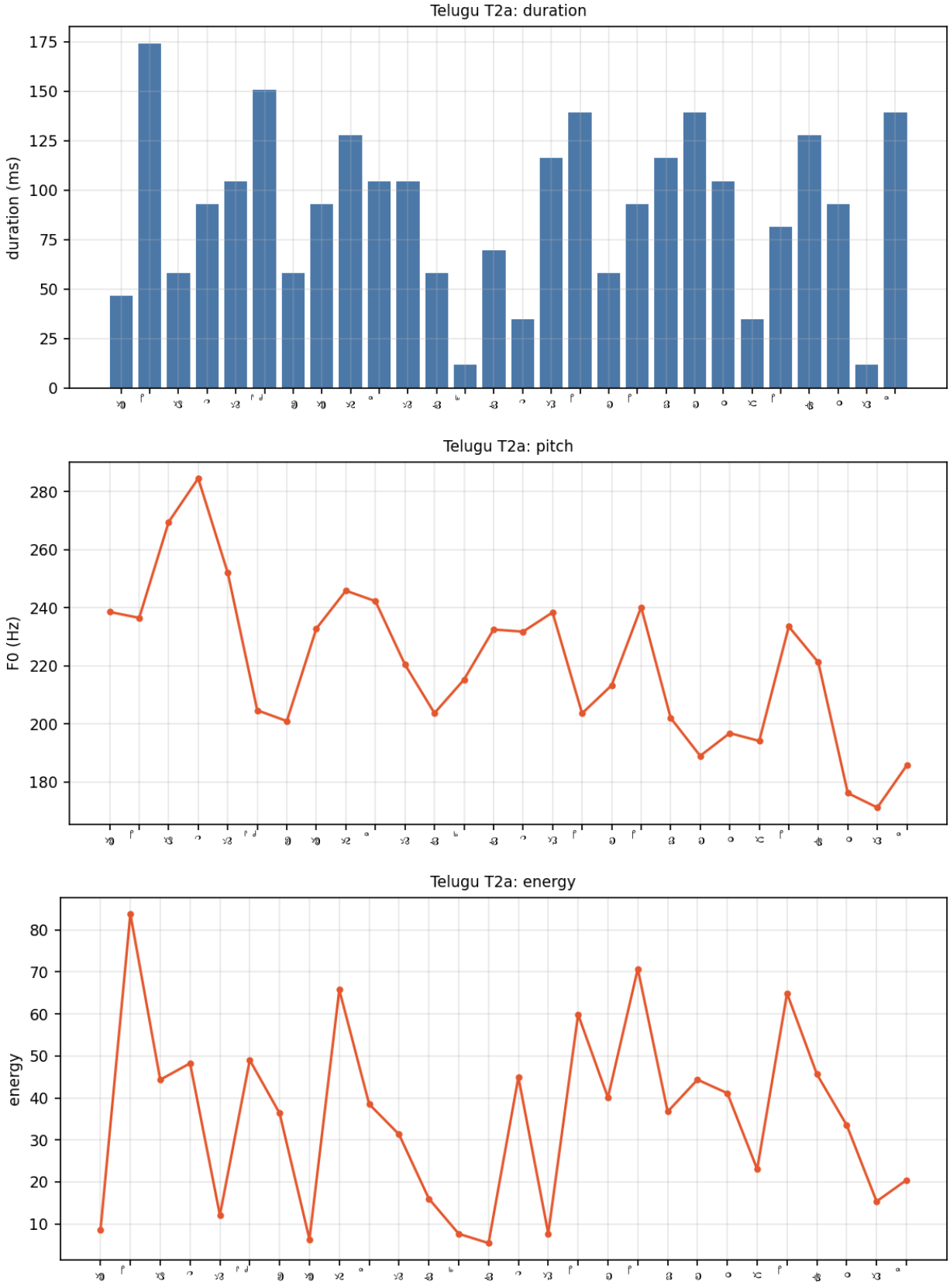


Figure 8: Predicted duration, pitch and energy for T2a.

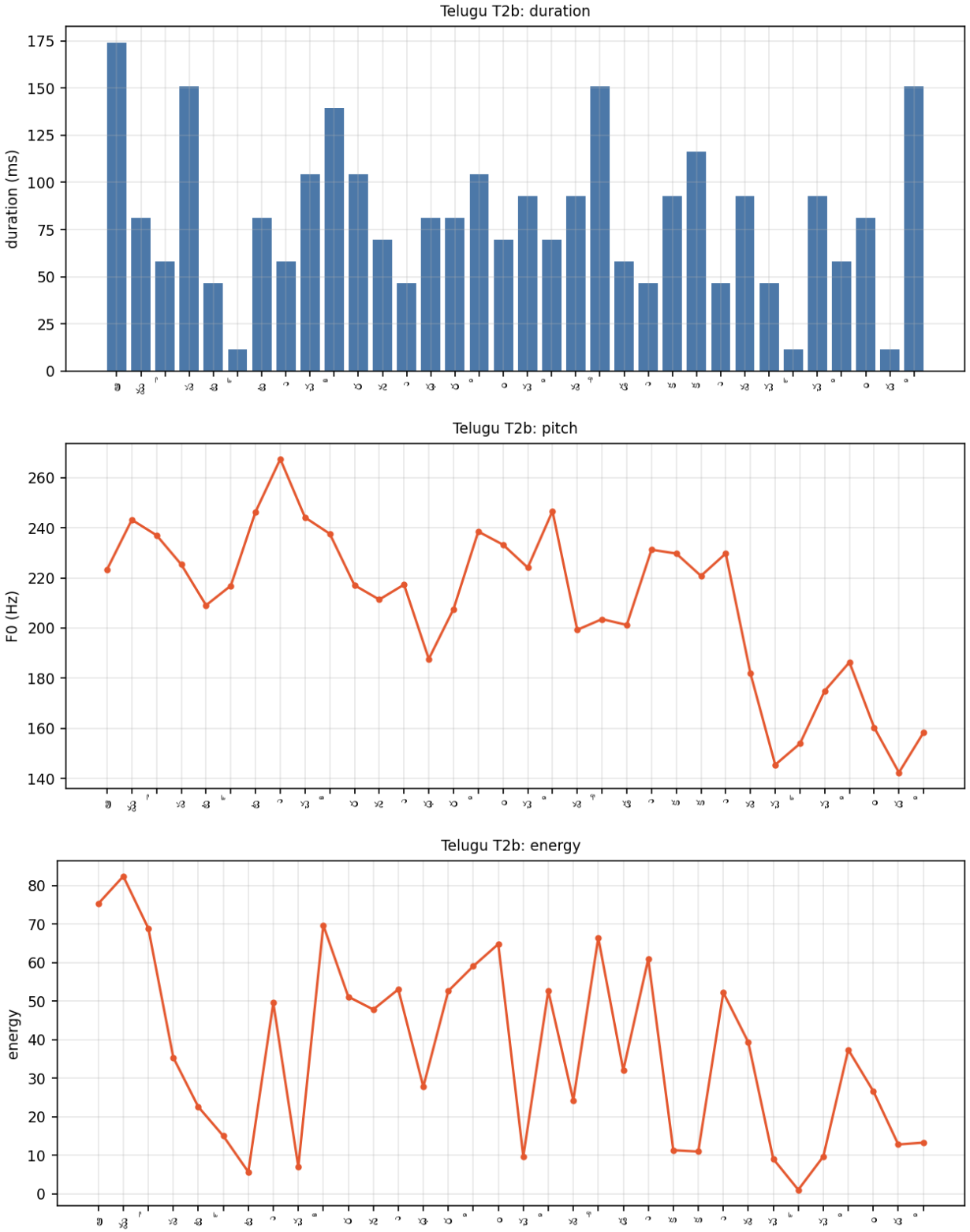


Figure 9: Predicted duration, pitch and energy for T2b.

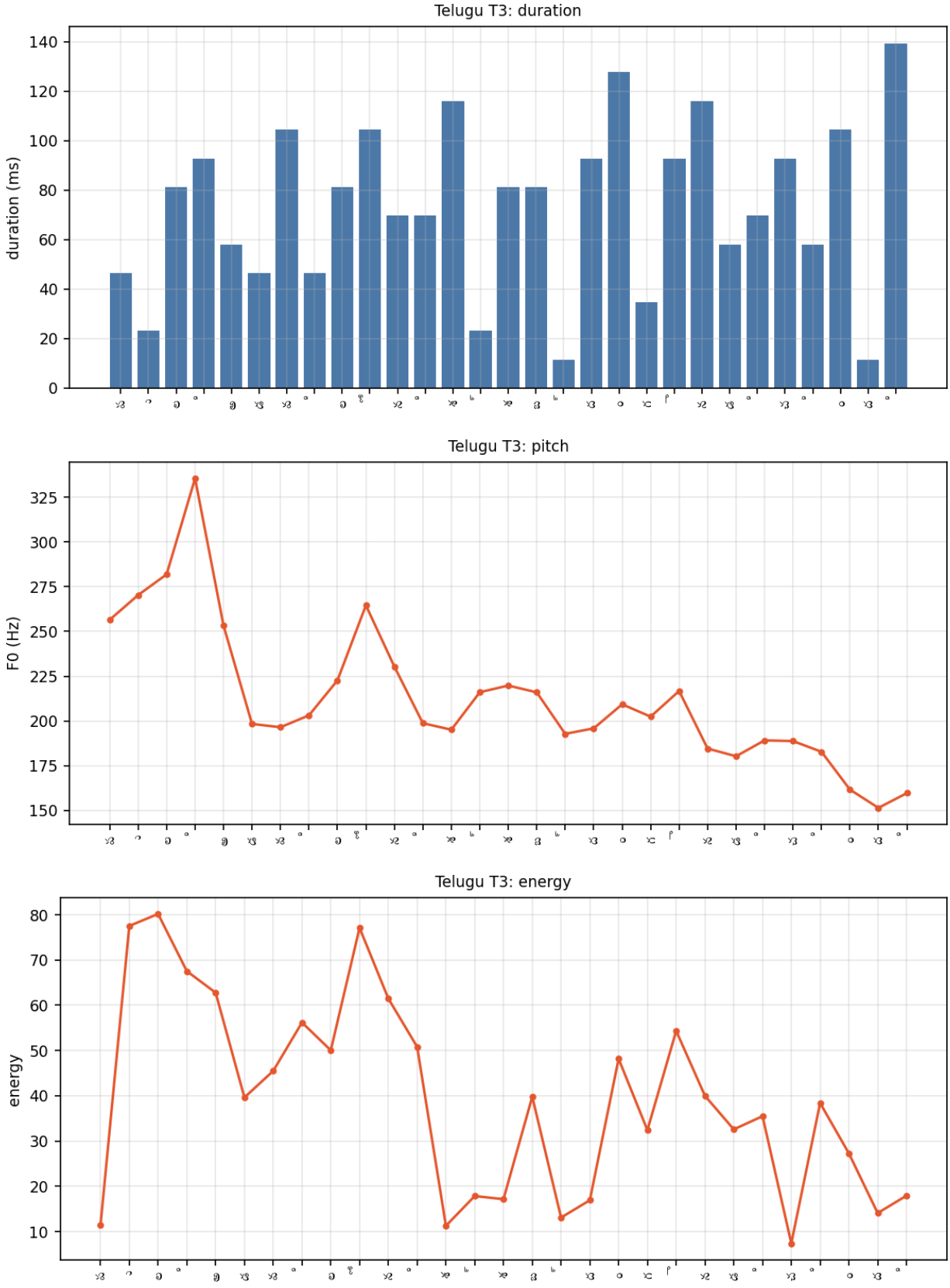


Figure 10: Predicted duration, pitch and energy for T3.

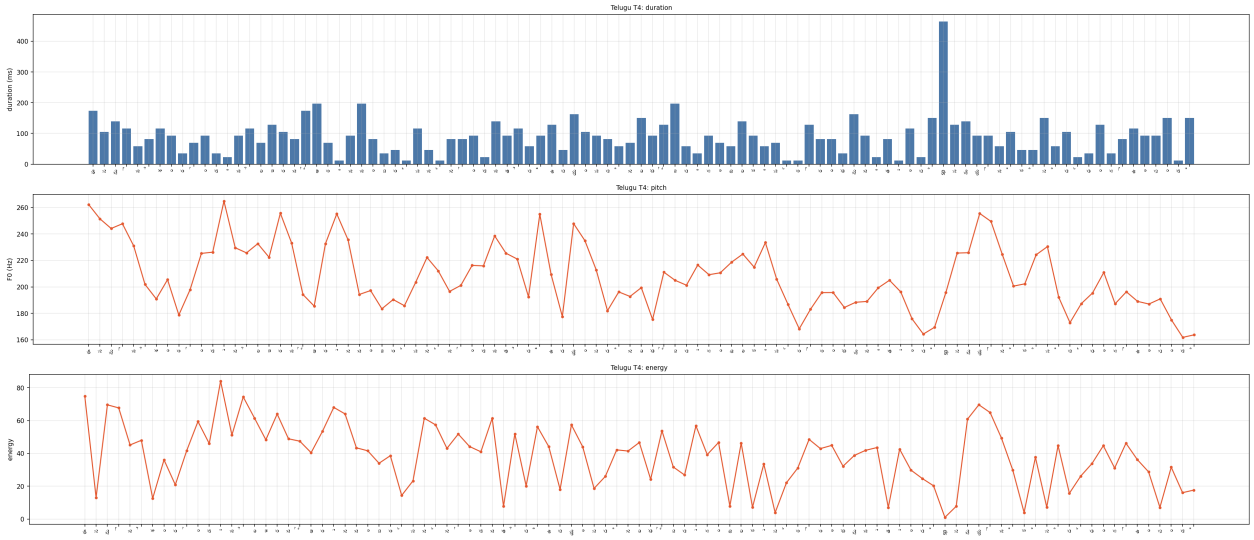


Figure 11: Predicted duration, pitch and energy for T4.

6 English versus Telugu

Table 5: Comparison of the two models.

	English	Telugu
Data	Svarah, noisy multi-speaker ASR corpus treated as one speaker	IndicTTS, studio recordings, two speakers
Symbols	ARPAbet phonemes from lexicon and g2p	Telugu characters (grapheme)
Alignment	pretrained MFA model	MFA trained from scratch (10+10 iterations)
Final val. loss	about 3.95	about 2.23
Train–val gap	about 2.9	about 1.4
Mean F0	176.5 Hz	192.8 Hz
Intelligibility of test sentences	all five clear	all five clear
Main weakness	homograph (*conduct*) not disambiguated	no speaker/pitch control tested; weaker alignment

Telugu reaches a much lower loss, although both systems produced intelligible speech for the test sentences. The Telugu corpus is clean and recorded for TTS, and its writing system is nearly phonetic, so a character-level model without a pronunciation dictionary is enough and the variance targets are easy to predict from text. The English data come from an ASR benchmark with many different speakers, microphones and accents, which we treated as one voice. This makes pitch and energy targets inconsistent and explains the large validation gap. In addition, English spelling is irregular and needs a lexicon plus g2p, which can fail on rare or ambiguous words.

6.1 Limitations and failures seen on the test sentences

- **Homographs (E2a/E2b):** the English front-end gives the same pronunciation for the noun and the verb *conduct*, so the intended stress contrast is not produced. In Telugu T2a/T2b the word *pattu* is pronounced the same in both meanings, so the pair does not test pronunciation but only context.
- **Single-speaker assumption for English:** Svarah has no speaker label in the version we used, so speaker variability is mixed into one voice, which probably hurts prosody and validation loss.
- **Alignment quality:** 35 English and 5 Telugu utterances failed to align, and the Telugu aligner was trained for only 10+10 iterations because of the 12-hour compute limit.
- **Telugu start-of-utterance pitch:** very high F0 values (up to about 300–335 Hz) in T1 and T3 may indicate an overshoot of the pitch predictor.
- **Single run and single listener:** one training run per language, no hyper-parameter search, and quality notes from one listener; no objective metric (for example MOS or WER of an ASR system) was computed.

6.2 Possible improvements

- Add part-of-speech or a heteronym dictionary to the English front-end so that the verb *conduct* gets second-syllable stress.

- Use a real single-speaker English TTS corpus, or diarise/cluster Svarah and train with a speaker embedding.
- Run the Telugu aligner with the full default schedule and filter poorly aligned utterances; evaluate with an ASR-based intelligibility score.
- Save checkpoints every 1,000 steps and select the one with the lowest validation loss, and use more training data or stronger regularisation for English.
- Render the Telugu symbol labels in the variance plots with a font that supports Telugu.

7 Reproducibility

7.1 Environment

Python 3.13 with the packages in `requirements.txt`: `torch`, `numpy`, `scipy`, `matplotlib`, `librosa`, `pyyaml`, `unidecode`, `inflect`, `g2p_en`, `nltk`, `pypinyin`, `tensorboard`, `tqdm`, `soundfile`, `pillow`, `tgt`, `pyworld` and `setuptools<81`. MFA 3.4.3 was installed with micromamba (`montreal-forced-aligner`) in a separate environment.

7.2 Commands

Install and run inference (CPU or GPU, with the provided checkpoints; tested on a fresh clone without a GPU):

```
git clone https://github.com/rashiii02/fastspeech2-english-telugu
cd fastspeech2-english-telugu
pip install -r requirements.txt
cd English && python inference.py --ckpt checkpoints/English_20000_fp16.pth.tar
cd ../Telugu && python inference.py --ckpt checkpoints/Telugu_20000_fp16.pth.tar
```

Audio and plots are written to `Results/<Language>/`. Other options are `-speaker` and `-out`.

Full pipeline from scratch (Svarah is gated on Hugging Face, so `HF_TOKEN` is required; MFA 3.4.3 must be on the PATH):

```
python scripts/pipeline.py data        # download, subsample (seed 42), resample
python scripts/pipeline.py align       # MFA alignment (Telugu: trained from scratch)
python scripts/pipeline.py textgrids   # place TextGrids for preprocessing
cd English && python preprocess.py preprocess.yaml && cd ..
cd Telugu  && python preprocess.py preprocess.yaml && cd ..
python scripts/pipeline.py split       # train / validation lists
cd English && python train.py -p preprocess.yaml -m model.yaml -t train.yaml
cd Telugu  && python train.py -p preprocess.yaml -m model.yaml -t train.yaml
```

Each language folder also contains a merged `config.yaml` with the preprocessing, model and training settings.

7.3 Repository layout

- `English/`, `Telugu/`: code, configuration and `checkpoints/`.
- `Results/English/`, `Results/Telugu/`: `training_loss.png`, `validation_loss.png`, `test_generated_audios/`, `variance_adaptor_plots/{duration,pitch,energy}/`, `frontend_log.json`.
- `Report/Technical_Report.pdf`, `README.md`, `requirements.txt`, `scripts/pipeline.py`.